Progress Update #3: Feature Encoding & Selection for Text-Only LLMs

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Outline

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The New Problem That Offline Extraction Creates

Once features are extracted offline, each feature must be *rendered textually* in the prompt. For every feature we face a design decision:

Temporal sequence vs. **Aggregated scalar statistic**

Amplitude is the critical case:

- MDS-UPDRS 3.4 explicitly references "*amplitude decrements near the end / midway / after the 1st tap*" — a temporal property.
- Whether scalar summaries *add value on top of* the sequence, or merely duplicate it, is not obvious a priori — this is what we test.

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Amplitude Sequence as a Feature

20 `tap_amp_norm` positions + 67 scalars = 87 features ($n=53$).

FS method	#sc	Amp seq?
Standard LASSO	16	4/20
Group LASSO	8	No
abess best-subset	2	No
Boruta	8	No
RFECV (RF)	7	1/20
Seq. Forward (RF)	5	3/20
Stability Sel.	5	2/20

Top inclusion freq.

Feature	Freq
amplitude_min	6/7
period_quartile_range	5/7
periodEntropy	4/7
amplitude_stdev	4/7
tap_amp_norm_10	3/7
tap_amp_norm_07	2/7
tap_amp_norm_11	2/7

4/7 methods pick amp-seq positions; late taps (07/10/11) dominate — matches MDS-UPDRS decrement criterion.

$\Rightarrow$ Most methods still *skip* amp-seq entirely — motivates Exp 2: is amp-seq informative even when FS leaves it out?

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Multi-Model Summary (Key Configurations)

Model	Condition	MAE↓	ACC↑	Kappa↑	AAcc↑
DeepSeek-R1 32B FS					
DS-R1-32b FS	with amp sequence	**0.49**	**0.53**	**0.57**	**0.98**
DS-R1-32b FS	no amp sequence	0.55	0.49	0.55	0.96
Doctor (Chien)	reference	0.36	0.64	0.83	1.00
Doctor (Tan)	reference	0.40	0.60	0.69	1.00

- Adding the amplitude sequence to the prompt: **MAE** -0.06**, ACC** $+0.04$**, Kappa** $+0.02$.
- The *with amp sequence* configuration approaches the doctor reference on MAE (0.49 vs. Tan 0.40 / Chien 0.36).

amp sequence: `tap_amplitudes_normalized` is rendered in the prompt as a per-tap list.

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Summary

- Extending Chen (2024) to text-only LLMs forced us to decide how to encode each feature in text.
- Exp 1 — Amp seq as FS candidate: 4/7 methods select late-tap positions (07/10/11), matching MDS-UPDRS decrement wording; 3/7 skip it entirely.
- Exp 2 — Adding the amp sequence to the prompt gives MAE -0.06, ACC $+0.04$; the sequence carries independent information.
- The *with amp sequence* configuration approaches the inter-rater doctor reference on MAE.

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